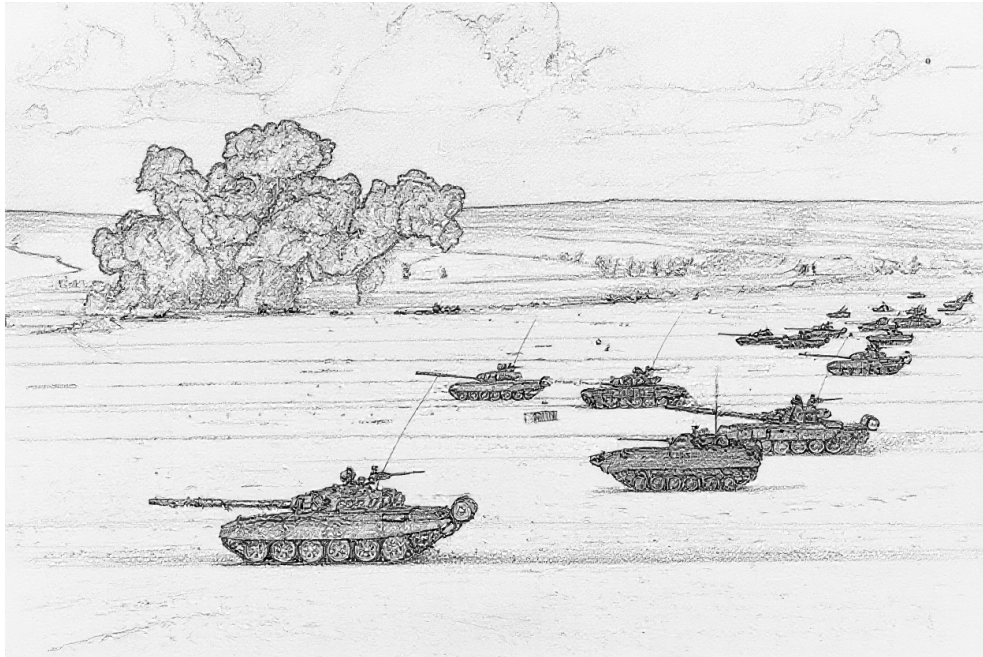


Air Power: Ground Unit and Ground Target Data

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INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:¹

“You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb.”

Here we deal with those “whatevers,” the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe’s extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

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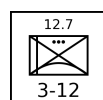
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. These units are presented below with their counters.

The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units.¹⁹

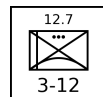
All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damage.

LIGHT AAA UNITS



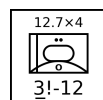
M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns.

The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.²⁰



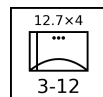
DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat, but here is used on a tall tripod suitable for air defense. A platoon has six guns.

The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms.²¹



M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles.

The M16 was used by United States forces during World War II and (primarily as an anti-personnel weapon) in the Korean War for airfield defense. Additionally, Israeli forces employed it during the 1956 Arab-Israeli War.²²



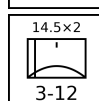
M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a towed carriage. A platoon has four gun mounts.

The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.²³



ZPU-1/2/4 Battery: The ZPU-1/2/4 have single, twin, or quad 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts.

The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.²⁴



Mobile ZPU-1/2/4 Section: This is a section of two light trucks mounting ZPU-1/2/4 guns.

Many users of the ZPU-1/2/4 mount it on a suitable truck for increased mobility. Like the towed versions, it has seen combat in many conflicts.²⁵



M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. The fire-control system has optical sights and a range-only radar. A platoon has four gun mounts.

The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.²⁶



M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. The fire-control system has optical sights and a range-only radar. A section has two vehicles.

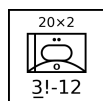
The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada.

From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a pair of FIM-92 squads.²⁷

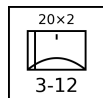


TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts.

The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.²⁸

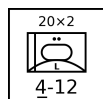


M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles. The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.²⁹



Twin Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts.

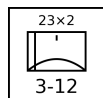
The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).³⁰



Panhard M3 DCA Section: The Panhard M3 DCA in *Air Strike* seems to correspond to Wikipedia M3 VDA or WeaponSystems M3 VDAA.

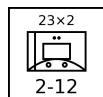
The Panhard M3 DCA vehicles have an armored turret for twin HS.820 20 mm guns mounted on the Panhard M3 APC. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. A section has two vehicles.

The Panhard M3 DCA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.³¹



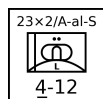
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. The fire-control system is entirely optical. A battery has six gun mounts.

The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.³²



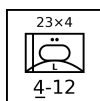
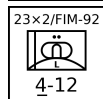
Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns.

As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility.³³



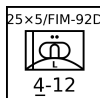
Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. A section has two vehicles. Each battery operates with one vehicle equipped with an RA 20S search and ranging radar.

The Sinai 23 entered service with the Egyptian Army in 1991.³⁴



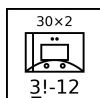
ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles.

The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.³⁵



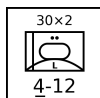
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. It does not have radar, but is equipped with a passive night IR sight. A section has two vehicles.

The LAV-AD entered service with the USMC in 1997, but was retired after only a few years.³⁶



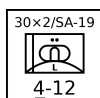
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles.

The M53/59 entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.³⁷



AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. It is an improved version of the AMX-30 DCA. A section has two vehicles.

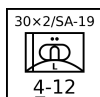
The AMX-30SA served only in the Saudi Army from 1979.³⁸



Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. It was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. A section has two vehicles.

The Tunguska entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

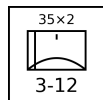
The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.³⁹



Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles.

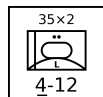
The Tunguska-M entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.⁴⁰

MEDIUM AAA UNITS



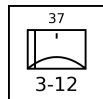
Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.)

The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).⁴¹



Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles.

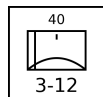
The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.⁴²



61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. A battery has six gun mounts.

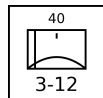
After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars.

The 61-K is referred to as the “M-38” in *Air Strike, Eagles of the Gulf*, and *The Speed of Heat*.⁴³



Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. A battery has six gun mounts.

It entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.⁴⁴



Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. A battery has six gun mounts.

NATO forces and others adopted it widely starting in 1952. Israel used batteries with the Super Fledermaus FCR for point-defense in the 1967 war.⁴⁵



Bofors L/70 BOFI and BOFI-R Battery: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. The BOFI-R version also adds an integrated fire-control radar. A battery has six gun mounts.

The BOFI is used by the armed forces of Brazil.⁴⁶



S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. A battery has six gun mounts.

It entered service in 1950 with Soviet forces as a replacement for the 61-K 37 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.⁴⁷



ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. The cannons are similar to the S-60 cannons. The fire-control system is optical. A section has two vehicles.

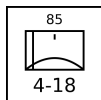
The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.⁴⁸



M51 Battery: The M51 Skysweeper has a single 75 mm gun in an open mount in a towed carriage. The mount incorporates the T38 radar fire-control system. Its rate of fire is 45 rounds per minute.

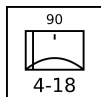
The M51 entered service with the US Army in 1951 and was used for defense of strategic locations in the USA and in Europe, but was retired in 1959. It also served with Greece, Japan, and Turkey, with the last examples being retired until the mid-1970s.⁴⁹

HEAVY AAA UNITS



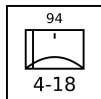
KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. The rate of fire is 10 rounds per minute. The fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.

It entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.⁵⁰



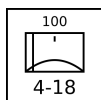
90 mm M2 Battery: The 90 mm M2 has a single 90 mm gun on a towed carriage. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery of four guns. The mechanical loading system permits a rate of fire of 24 rounds per minute.

The M2 entered service with the US Army in 1943, following on from the similar M1 in 1940, and served as the main US heavy AA gun in WW2 and the Korean War. It was retired in the mid 1950s. It also served with Canadian (until 1960), French, and Turkish forces.⁵¹



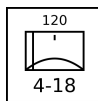
QF 3.7 inch Battery: The QF 3.7 inch has a single 94 mm gun on a towed carriage, but typically is used from a prepared firing site. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery. In the final Mk VI version, the mechanical loading system permits a rate of fire of 20 rounds per minute.

The gun entered service with the British Army in 1937, served as its principal heavy AA gun during WW2 and beyond, and was withdrawn in 1957. It also served with Australia, Belgium, Canada, Sri Lanka, Cyprus, India, Ireland, Israel, Malta, Nepal, New Zealand, Pakistan, South Africa, Yugoslavia, and Portugal.⁵²



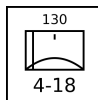
KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.

It entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.⁵³



120 mm M1 Battery: The M1 has a single 120 mm gun in an open mount on a towed carriage, but typically is used from a prepared firing site. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery of four guns firing proximity-fuzed shells. The rate of fire is 12 rounds per minute.

The M1 entered service with the US Army in 1944, and though deployed to the Pacific Theater it did not engage on enemy aircraft. After the war, it was widely deployed to defend strategic targets against Soviet Tu-4 bombers, but was withdrawn by 1960. It also served with Canadian forces.⁵⁴



KS-30 Battery: The KS-30 has a single 130 mm gun in an open mount on a towed carriage. Developed started after WW2, and it has features adopted from captured German guns as well as British QF 3.7 inch guns. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The rate of fire is 12 rounds per minute. A battery has six gun mounts.

It entered service in 1955 with Soviet forces, was used to defend strategic sites, and served until 1962. It was not widely exported, but also served with the North Vietnamese forces in the Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.⁵⁵

AAA Units Table

Type	Size ^a	Year	Defense Strength		Sighting Range	Mobility	VPs		Gun	AAA Class	Altitude	Range			Hit			Damage Rating	FCR	Night IR Sights		SAM
							3D/2D/D					S	M	L	S	M	L					
M2	Platoon	1933	3	12	U		3/2/1		12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—	
DShK-38	Platoon	1938	3	12	U		3/2/1		12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—	
M16	Section	1944	3	12	U		6/3		12.7 mm × 4	L	5	2	3	4	4	3	2	2	—	—	—	
M55	Platoon	1945	3	12	T		4/3/1		12.7 mm × 4	L	5	2	3	4	5	4	3	2	—	—	—	
ZPU-1	Battery	1949	3	12	T		3/2/1		14.5 mm	L	6	2	3	5	3	2	1	1	—	—	—	
ZPU-2	Battery	1949	3	12	T		4/3/1		14.5 mm × 2	L	6	2	3	5	4	3	2	1	—	—	—	
ZPU-4	Battery	1949	3	12	T		5/3/2		14.5 mm × 4	L	6	2	3	5	5	4	3	2	—	—	—	
Mobile ZPU-1	Section	1949	2	12	G/P		2/1		14.5 mm	L	6	2	3	5	2	1	0	1	—	—	—	
Mobile ZPU-2	Section	1949	2	12	G/P		3/2		14.5 mm × 2	L	6	2	3	5	3	2	1	1	—	—	—	
Mobile ZPU-4	Section	1949	2	12	G/P		4/2		14.5 mm × 4	L	6	2	3	5	4	3	2	2	—	—	—	
M167	Platoon	1967	3	12	T		7/5/2		20 mm × 6	L	8	2	4	6	6	5	4	3	R/HF	—	—	
M163	Section	1968	4	12	U		8/4		20 mm × 6	L	8	2	4	6	5	4	3	3	R/HF	—	—	
TCM-20	Platoon	1970	3	12	T		5/3/2		20 mm × 2	L	8	2	4	6	4	4	3	2	—	—	—	
M3 TCM-20	Section	1970	3	12	U		6/3		20 mm × 2	L	8	2	4	6	3	3	2	2	—	—	—	
Single Rh-202	Battery	1972	3	12	T		4/3/1		20 mm	L	8	2	4	6	4	3	2	2	—	—	—	
Twin Rh-202	Battery	1972	3	12	T		5/3/2		20 mm × 2	L	8	2	4	6	5	4	3	3	—	—	—	
Panhard M3 DCA	Section	1975	4	12	U		7/4		20 mm × 2	L	8	2	4	6	4	4	3	2	R/UF	—	—	
ZU-23	Battery	1960	3	12	T		6/4/2		23 mm × 2	L	9	2	5	8	5	4	3	3	—	—	—	
Mobile ZU-23	Section	1960	2	12	G/P		5/3		23 mm × 2	L	9	2	5	8	4	3	2	3	—	—	—	
Sinai 23 Ayn-al-Saqr	Section	1992	4	12	U		10/5		23 mm × 2	L	9	2	5	8	5	4	3	3	R/UF	—	Ayn-al-Saqr	
Sinai 23 Stinger	Section	1992	4	12	U		12/6		23 mm × 2	L	9	2	5	8	5	4	3	3	R/UF	—	FIM-92	
ZSU-23-4	Section	1965	4	12	U		9/5		23 mm × 4	L	9	2	5	8	5	4	3	3	W/VF	—	—	
LAV-AD	Section	1997	4	12	U		14/7		25 mm × 5	L	10	3	6	9	5	4	3	4	—	Y	FIM-92D	
M53/59	Section	1959	3	12	G/P		7/4		30 mm × 2	L	11	4	7	9	3	2	1	3	—	—	—	
AMX-30SA	Section	1979	4	12	U		10/5		30 mm × 2	L	11	4	7	9	5	4	3	3	W/VF	—	—	
Tunguska	Section	1984	4	12	U		20/10		30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19	
Tunguska-M	Section	1990	4	12	U		22/11		30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19	
Oerlikon GDF	Battery	1963	3	12	T		7/5/2		35 mm × 2	M	12	4	8	10	4	3	2	4	—	—	—	
Oerlikon GDF	Section	1963	3	12	T		5/2		35 mm × 2	M	12	4	8	10	3	2	1	4	—	—	—	
Gepard	Section	1976	4	12	U		11/6		35 mm × 2	M	12	4	8	10	5	4	3	4	W/VF	—	—	
61-K	Battery	1939	3	12	T		5/3/2		37 mm	M	12	4	8	11	3	2	1	4	—	—	—	
Bofors L/60	Battery	1935	3	12	T		5/3/2		40 mm	M	15	4	8	12	2	2	1	4	—	—	—	
Bofors L/70	Battery	1952	3	12	T		6/4/2		40 mm	M	15	4	8	12	3	3	2	4	—	—	—	
Bofors L/70 BOFI	Battery		3	12	T		8/5/3		40 mm	M	15	4	8	12	5	5	4	4	—	—	—	
Bofors L/70 BOFI-R	Battery		3	12	T		9/6/3		40 mm	M	15	4	8	12	6	6	5	4	W/VF	—	—	
S-60	Battery	1950	3	12	T		8/5/3		57 mm	M	18	5	10	15	2	2	1	5	—	—	—	
ZSU-57-2	Section	1955	4	12	U		6/3		57 mm × 2	M	18	5	10	15	2	1	1	5	—	—	—	
M51	Battery	1953	4	18	T		9/6/3		75 mm	M	30	6	12	18	4	4	3	6	—	—	—	
KS-12	Battery	1939	4	18	T		9/6/3		85 mm	H	27	6	12	18	2	1	0	6	—	—	—	
90 mm M2	Battery	1939	4	18	T		9/6/3		90 mm	H	43	7	14	21	3	2	1	6	—	—	—	
QF 3.7 Inch	Battery	1939	4	18	T		9/6/3		94 mm	H	45	9	18	27	3	2	1	6	—	—	—	
KS-19	Battery	1948	4	18	T		10/7/3		100 mm	H	39	7	14	21	1	1	0	6	—	—	—	
120 mm M1	Battery	1944	4	18	T		9/6/3		120 mm	H	57	11	22	33	2	2	1	7	—	—	—	
KS-30	Battery	1955	4	18	T		11/7/4		130 mm	H	64	12	24	36	1	1	0	7	—	—	—	

^a Platoons and batteries are available as sections with corresponding modifications to their damage resiliences and hit rolls.

NOTES

1. Robin Olds

The full quotation contains a second insight: “You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren’t there to win it anyway, so it was all a waste of time.” (Sherwood, John. *Fast Movers: Jet Pilots and the Vietnam Experience*, chapter 1)

19. AAA Values

This long note summarizes how I have determined the AAA values for AAA units. VP values are discussed separately.

I compare AAA values for AAA units from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* to the actual characteristics of the guns they represent. My aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns. That said, I do comment on the model.

AAA Values for Batteries: I start by attempting to understand the AAA values for batteries.

The upper part of the following table shows the original data for gun batteries from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
Original									
M2/DShK-38	5	2/3/4	3/2/1	1	—	12.7	500	890	AS
M55	5	2/3/4	5/4/3	2	—	12.7	2000	890	EOTG
ZPU-1	6	2/3/5	3/2/1	1	—	14.5	600	1000	EOTG/TSOH
ZPU-2	6	2/3/5	4/3/2	1	—	14.5	1200	1000	AS
ZPU-4	6	2/3/5	5/4/3	2	—	14.5	2400	1000	AS
ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	TSOH
TCM-20	8	2/4/6	4/4/3	2	—	20	1450	860	EOTG
Single Rh-202	8	2/4/6	3/2/1	2	—	20	1030	1044	AS
Twin Rh-202	8	2/4/6	5/4/3	2	—	20	2060	1044	AS
M167	8	2/4/6	6/5/4	3	R	20	3000	1030	AS
ZU-23	9	2/5/8	4/3/2	2	—	23	2000	980	AS/TSOH
Gemag	10	3/6/9	6/5/4	4	R	25	1800	1040	AS
Oerlikon GDF	12	4/8/10	5/4/3	4	—	35	1100	1175	AS
61-K (M-38)	12	4/8/11	3/2/1	4	—	37	165	880	AS/TSOH
Bofors L/60	15	4/8/12	2/2/1	4	—	40	135	865	EOTG
Bofors L/70	15	4/8/12	3/3/2	4	—	40	330	1000	AS
BOFI	15	4/8/12	4/4/3	4	W	40	330	1000	AS
S-60	18	5/10/15	2/2/1	5	—	57	110	1000	AS/TSOH
KS-12	27	6/12/18	2/1/0	6	—	85	10	790	AS/TSOH
KS-19	39	7/14/21	1/1/0	6	—	100	15	900	AS
Modified									
ZPU-4		5/4/3				— for rate of fire			
Single Rh-202		4/3/2				— for rate of fire			
Twin Rh-202			3			— for rate of fire			
ZU-23		5/4/3	3			— for rate of fire			
Oerlikon GDF		4/3/2				— for rate of fire			
BOFI		5/5/3				— for FCR-W			
New									
M51	30	6/12/18	4/4/3	6	W	75	45	854	
90 mm M2	32	7/14/21	3/2/1	6	—	90	24	820	
QF 3.7 inch Mk VI	45	9/18/27	3/2/1	6	—	94	20	1044	
120 mm M1	57	11/22/33	2/2/1	7	—	120	12	945	
KS-30	64	12/24/36	1/1/0	7	—	130	12	970	

Qualitatively, altitude and range increase with caliber, hit rolls increase with rate of fire, and damage rating increases with both caliber and rate of fire. This is as expected.

Quantitatively:

- Approximately doubling the rate of fire increases the hit rolls by one, with typical values of 3/2/1 around 500 rounds per minute, 4/3/2 around 1000, and 5/3/2 around 2000 and above.

- The base damage rating is 1 for 12.7–14.5 mm, 2 for 20–23 mm, 3 for 25–30 mm, 4 for 35–40 mm, 5 for 57 mm, and 6 for 85–100 mm.
- The damage rating is increased by one if the rate of fire is 1800 rounds per minute or more.

Most of the original values can be used as they are, but a few of the hit rolls need adjusting:

- ZPU-4 and ZU-23

There are six AAA batteries with rates of fire per mount of about 2000 rounds per minute or more: the M55, ZPU-4, twin Rh-202, M167, ZU-23, and Gemag. After accounting for the +1 modifier for the FCR-R of the M167 and Gemag, all have hit rolls of 5/4/3 except for the ZPU-4 from *The Speed of Heat* (although the one from *Air Strike* does have 5/4/3) and the ZU-23.

Furthermore, all of these except for the ZU-23 (both *Air Strike* and *The Speed of Heat* versions) have damage ratings increased by one above their baseline values.

It seems that either there is some reason to reduce the effectiveness of the ZPU-4 and ZU-23 compared to these other guns, or perhaps they have inconsistent values rolls.

The ZU-23 is undeniably basic, and one might think that would be reason to reduce its effectiveness. However, the TCM-20 is basic too and has both a lower rate of fire than the ZU-23 (1450 compared to 2000) and a smaller caliber (20 mm compared to 23 mm), yet its hit rolls are higher than the ZU-23 in *The Speed of Heat* and its damage rating is the same.

The best solution would seem to be to give these guns the values we would expect from comparing them to other guns with similar rate of fire and caliber. Therefore, I give both the ZPU-4 and ZU-23 hit rolls of 5/4/3 and the ZU-23 a damage rating of 3.

(I also considered the idea that increasing the rate of fire increases *either* the hit rolls *or* the damage rating. However, while this might explain the ZPU-4 from *The Speed of Heat* and the twin Rh-202, it does not correctly predict the values of the M167 or Gemag.)

- Single RH-202

The single Rh-202 has unexpectedly low original hit rolls of 3/2/1 for its rate of fire. The values of 4/3/2 for the ZPU-2 and TCM-20, which have a similar rate of fire, are probably more appropriate.

- Twin Rh-202

The twin has an unexpectedly low original damage rating of 2 for its rate of fire. A value of 3, increased by one from the baseline for the caliber, is probably more appropriate.

- Oerlikon GDF

The Oerlikon GDF has unexpectedly high original hit rolls of 5/4/3 for its rate of fire. The values of 4/3/2 for the ZPU-2, which has a similar rate of fire, are probably more appropriate.

- BOFI

The BOFI has unexpectedly low original hit rolls of 4/4/3. Values of 5/5/3, obtained by applying a +2 modifier for FCR-W to the Bofors L/70, are probably more appropriate. (Note that the “BOFI” in *Air Strike* corresponds to what I call the “BOFI-R” below.)

I adopt these modified values.

Also, I note the following:

- FCR-R

The M167 and Gemag have hit rolls of 6/5/4, and the twin Rh-202 has 5/4/3. Otherwise, they are similar, except that the M167 and Gemag have FCR-R, and the Rh-202 does not (and the M167 has a moderate advantage in rate of fire). I consider the FCR-R of the M167 and Gemag to be largely responsible for increasing their hit rolls by +1 compared to the Rh-202.

- Bofors

The hit rolls of 2/2/1 for the Bofors L/60, 3/2/1 for the 61-K (M-38), and 3/3/2 for the Bofors L/70 seem to reflect the increase in rate of fire.

The L/60 and L/70 have the same maximum altitude of 16 and maximum range of 12, despite the L/60 having lower muzzle velocity. However, I am not especially motivated to attempt to change this, since the Bofors is primarily a weapon for point-defense against aircraft up to about level 10.

The real BOFI and BOFI-R also have proximity fuzes, and I consider these will increase their hit rolls by +1.

AAA Values for Sections: I now turn to sections. Again, my aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns.

The upper part of the following table shows the original data for gun sections from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). Again, I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
Original									
M16	5	2/3/4	4/3/2	2	—	12.7	2000	890	EOTG
Mobile ZPU-4	6	2/3/5	3/3/2	2	—	14.5	2400	1000	AS
M163	8	2/4/6	5/4/3	3	R	20	3000	1030	AS
M3 TCM-20	8	2/4/6	3/3/2	2	—	20	1450	860	EOTG
Panhard	8	2/4/6	2/2/1	2	—	20	1450	860	AS
Mobile ZU-23	9	2/5/8	2/2/1	2	—	23	2000	980	AS
Nile 23	9	2/5/8	3/3/2	2	—	23	2000	980	EOTG
ZSU-23-4	9	2/5/8	5/4/3	3	W	23	4000	980	AS
Blazer	10	3/6/9	5/4/3	4	W	25	1800	1040	AS
M53/59	11	4/7/9	4/3/2	3	—	30	1000	1000	AS
AMX-30	11	4/7/9	5/4/3	3	W	30	1200	970	AS
ZSU-30-2	11	4/7/9	5/4/3	4	W	30	5000	960	AS
Gepard	12	4/8/10	5/4/3	4	W	35	1100	1175	AS
ZSU-57-2	18	5/10/15	2/1/1	5	—	57	240	1000	EOTG
Modified									
Mobile ZPU-4			4/3/2			—	to match the ZPU-4		
Panhard			3/3/2			—	to match the M3 TCM-20		
Mobile ZU-23			4/3/2	3		—	to match the ZU-23		
Nile 23			4/3/2	3		—	to match the ZU-23		
M53/59			3/2/1			—	to match the AMX-30		

Comparing similar weapons, batteries and sections have the same maximum altitudes, ranges, and damage ratings. However, batteries (with four to six mounts) typically have a +1 advantage on their hit rolls compared to sections (with two mounts). In a very few cases, the difference is 0, and in a very few others, it is +2. A difference of 1 in the hit rolls is commensurate with the approximate doubling of the rate of fire from a section to a battery.

To determine the appropriate AAA for sections, the following recipe seems to work in most cases:

- Take the value for an equivalent battery without FCR and apply an adjustment of –1 to the hit rolls to account for the reduced rate of fire in the section. (Only do this if the rate of fire is reduced; see the comment on the ZSU-23-4 and ZSU-57-2 below.)
- Apply an adjustment of +1 to the hit rolls for FCR-R or similar.
- Apply an adjustment of +2 to the hit rolls for FCR-W.
- The preceding steps would then predict hit rolls of 6/5/4 for the Blazer and 7/6/5 for the ZSU-23-4 and ZSU-30-2. To conform to the original values, I limit the hit rolls to 5/4/3. It would appear that in the underlying model, beyond a certain rate of fire, other factors take over to limit the probability of a hit.

Again, most of the original data can be used as they are, but a few of the values are worth commenting upon or need adjusting.

- **Agreements**
The original hit rolls and the predicted hit rolls are in agreement for the M16 (from the M55 battery), M3 TCM-20 (from the TCM-20 battery), M163 (from the M167 battery), the Blazer (from the Gemag), and the Gepard (from the Oerlikon GDF).
- **Mobile ZPU-4**
The original hit rolls for the mobile ZPU-4 section are 3/3/2, but 4/3/2 is predicted from the values of 5/4/3 for the battery and is in better agreement with the values for the M16.
- **Panhard M3 DCA**
The Panhard M3 DCA has unexpectedly low original hit rolls of 2/2/1 compared to the TCM-20 battery and the M3 TCM-20 section, despite having very similar guns. Values of 3/3/2 are probably more appropriate.

- **Mobile ZU-23 and Nile 23**

The original hit rolls for the mobile ZU-23 and Nile 23 section are 2/2/1 and 3/3/2, despite them being quite similar. The predicted values of 4/3/2 are probably more appropriate and are in better agreement with the values for the M16. Also, as with the ZU-23 battery, a damage rating of 3 is probably more appropriate given their high rate of fire.

- **M53/59**

The original hit rolls for the M53/59 section are 4/3/2. The predicted values of 3/2/1, derived from the modified hit rolls of 4/3/2 for the Oerlikon GDF battery and also from the AMX-30SA after discounting the FCR-W, are probably more appropriate.

- **ZSU-57-2**

The ZSU-57-2 section has the same hit rolls as the S-60 battery. This is probably appropriate, as the ZSU-57-2 is a twin mount and this largely compensates for the reduced rate of fire of the section. The same applies to the quad ZSU-23-4 compared to the twin ZU-23.

I adopt these modified values.

Also, I note:

- **Panhard M3 DCA**

The Panhard M3 DCA has an FCR-R capability, which is discussed below but not considered here.

- **AMX-30 DCA**

The real AMX-30 DCA does not have an FCR. The AMX-30 DCA in the game seems to correspond to the similar AMX-30SA, which does.

The AAA values for new guns are discussed when they are present.

Summary: I have developed an empirical understanding of the model for AAA values. Consideration of outliers suggests:

- Increasing in the hit rolls of the mounted ZPU-4, single Rh-202, Panhard M3 VDA, ZU-23 (and derivatives), and BOFI.
- Decreasing the hit rolls of the Oerlikon GDF.
- Increasing the damage rating of the twin Rh-202 and the ZU-23 guns (and derivatives).

I have adopted these changes.

20. M2 Platoon

The AAA values are from *Air Strike*.

Sources:

- Wikipedia on the M2.
- WeaponSystems on the M2.

21. DShK-38 Platoon

The AAA values are from *Air Strike*.

Sources:

- Wikipedia on the DShK.
- WeaponSystems on the DShK.

22. M16 Section

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M16, M45, and M2.
- WeaponSystems on the M16 and M2.

23. M55 Battery

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M45 and M2.
- WeaponSystems on the M2.

24. ZPU-1/2/4 Battery

The AAA values for the ZPU-1 are from *Eagles of the Gulf* and *The Speed of Heat*. The AAA values for the ZPU-2 are from *Air Strike*. The AAA values for the ZPU-4 are from *Air Strike* rather than *The Speed of Heat*, for the reasons described in the extended note above.

Sources:

- Wikipedia on the ZPU.
- WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

25. Mobile ZPU-1/2/4 Section

AAA values for the mobile ZPU-4 are from *Air Strike*, except that I have adopted hit rolls of 4/3/2 rather than 3/3/2 as described in the extended note

above. The mobile ZPU-1/2 do not appear in *Air Strike*, *Eagles of the Gulf*, or *The Speed of Heat*, but they are very common. As described in the extended note above, I have produced AAA hit values for the sections from those of the ZPU-1 and ZPU-2 batteries.

Sources:

- Wikipedia on the ZPU.
- WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

26. M167 Platoon

The AAA values are from *Air Strike*.

The 1976 Operator's Manual describes the use of an AN/TVS-2B or AN/TVS-5 night sight. These are first- and second-generation image intensifiers. They are mounted next to the tracking sight and appear to be useful only for ground targets.

Sources:

- Wikipedia on the M167 and M61 Vulcan.
- WeaponSystems on the M167 and M61.
- Department of the Army, TM 9-1005-286-10, 1976: "Operator's Manual (Crew) for Gun, Air Defense Artillery, Towed".

27. M163 Section

The AAA values are from *Air Strike*.

The 1976 Operator's Manual describes the use of an AN/VPS-2 range-only radar.

The 1976 Operator's Manual describes the use of an AN/TVS-2B night sight. This is a first-generation image intensifier. It is mounted next to the tracking sight and appears to be useful only for ground targets.

The PIVADS version introduced in 1984 had an improved sight, but

Sources:

- Wikipedia on the M163, M167, and M61 Vulcan.
- WeaponSystems on the M163, M167 and M61.
- Department of the Army, TM 9-2350-300-10, 1976: "Operator's Manual For Gun, Air Defense Artillery, Self-Propelled 20-mm, M163".

28. TCM-20 Platoon

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the TCM-20 and HS.404.
- WeaponSystems on the TCM-20.

29. M3 TCM-20 Section

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M3 TCM-20, TCM-20, HS.404.
- WeaponSystems on the M3 TCM-20 and TCM-20.

30. Rh-202 Battery

The AAA values are from *Air Strike*, with the hit values for the single mount changed from 3/2/1 to 4/3/2 and the damage value for the twin mount changed from 2 to 3, as described in the extended note above.

Air Strike has single and twin Rh-202 batteries. Single mounts seem to be used for AFVs (e.g., Marder, Luchs, and Wiesel) and naval applications, but all AAA mounts seem to be twin.

Sources:

- Wikipedia on Rh-202.
- WeaponSystems on the Rh-202 twin mount and Rh-202 gun.

31. Panhard M3 DCA Section

The AAA data are from *Air Strike*, but modified as described in the extended note above to give hit rolls of 3/3/2 without FCR. However, the Panhard M3 DCA radar-equipped vehicle seems to be able to search and share speed and range information with the other vehicle in the section, which seems equivalent to FCR-R. Therefore, I have increased the hit rolls by one to give 4/4/3 with FCR-R.

Sources:

- Wikipedia on the Panhard M3 VDA and HS.820.
- WeaponSystems on the Panhard M3 VDAA.

32. ZU-23 Battery

The AAA values are from *Air Strike*, but the hit rolls are modified as from 4/3/2 to 5/3/2 and the damage rating from 2 to 3 as described in the extended note above.

Sources:

- Wikipedia on the ZU-23.
- WeaponSystems on the ZU-23 and 2A14.

33. Mobile ZU-23 Section

The AAA values are from *Air Strike* and *The Speed of Heat*, but the hit rolls of 4/3/2 were obtained from those for the ZU-23 battery using the recipe described above, and the damage was modified from 2 to 3 as described in the extended note above.

Sources:

- Wikipedia on the ZU-23.
- WeaponSystems on the ZU-23 and 2A14.

34. Sinai 23 Section

The gun properties are the same as the mobile ZU-23, but with +1 to the hit rolls for the FCR-R. This means that the Sinai 23 (with two barrels) then has the same hit rolls and damage rating as the ZSU-23-4 (with four barrels). Perhaps this is acceptable, since there are 30 years between these systems.

The turret is the Dassault TA-23E and seems to be similar to the LAV-AD turret, but with a pair of 23 mm guns instead of a GAU-12 25 mm gun. The AAA values were taken from the mobile ZU-23 battery. I don't know which model of Stinger was used; I would guess the FIM-92A.

The brochure explicitly mentions a "day sight"; there is no sign of an IR sight.

A platoon seems to consist of three "satellite" fire units and one "leader" unit with a RA 20S radar. The radar has only 1 degree precision, so it is not a traditional FCR, but seems to give warning, cueing, and range for optical tracking. I will treat it as an FCR-R and give a +1 modifier.

The radar works in the E-band. This is 2-3 GHz or UHF.

Perhaps the most accurate way to handle the radar would be to have a platoon of 3 fire units and 1 radar unit, and for each D determine if the radar unit is killed on a 3-.

Sources:

- Wikipedia on Sinai 23 and ZU-23.
- Brochure describing the Sinai 23.
- Description of RA 20S FCR.

35. ZSU-23-4 Section

The AAA values are from *Air Strike*.

Sources:

- Wikipedia on the ZSU-23-4.
- WeaponSystems on the ZSU-23-4 and 2A7.

36. LAV-AD Section

The system seems to be similar to the Blazer, but without radar and with a laser range finder and night IR sights. I've adopted the range and damage rating for the Gemag battery in *Air Strike* with the hit rolls reduced by one for being a section. The laser rangefinder substitutes for the FCR-R, so the hit rolls are not reduced further.

Sources:

- Wikipedia on the LAV-AD, GAU-12, and FIM-92.
- Dickerson, C.R., 1991: "LAV-AD: Mobile Air Defense for the Ground Maneuver Force".

37. M53/59 Section

The AAA values are from *Air Strike*, but the hit rolls were reduced from 4/3/2 to 3/2/1 as described in the extended note above. If the VADS is classed as armored, should this be armored too?

Sources:

- Wikipedia on the M53/59.
- WeaponSystems on the M53/59 and M53.

38. AMX-30SA Section

I adopt the *Air Strike* values given for the AMX-30 DCA. The real DCA does not have a tracking radar, but the *Air Strike* one does, matching the SA.

Sources:

- Wikipedia on the AMX-30DCA/SA and HS.831A.
- WeaponSystems on the AMX-30DCA HS.831A.

39. Tunguska Section

The AAA values are from *Air Strike* for the ZSU-30-2.

Sources:

- Wikipedia on the Tunguska.
- WeaponSystems on the Tunguska, 2A38, and 9M311.

40. Tunguska-M Section

The AAA values are from *Air Strike* for the ZSU-30-2.

Sources:

- Wikipedia on the Tunguska.
- WeaponSystems on the Tunguska, 2A38, and 9M311.

41. Oerlikon GDF Battery

The AAA values for the battery are from *Air Strike* with the hit rolls reduced from 5/4/3 to 4/3/2 as described in the extended note above.

Sources:

- Wikipedia on the GDF.
- WeaponSystems on the GDF and KDA.

42. Gepard Section

The AAA values are from *Air Strike*.

Sources:

- Wikipedia on the Gepard.
- WeaponSystems on the Gepard and KDA.

43. 61-K Battery

The AAA values are from *Air Strike* and *The Speed of Heat*.

I can find no sources other than *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* that refer to the 61-K 37 mm gun as the “M-38”. Another designation for the gun is “M-1939”, so I could understand “M-39”.

Sources:

- Wikipedia on the 61-K.
- WeaponSystems on the 61-K.

44. Bofors L/60 Battery

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the Bofors L/60.
- WeaponSystems on the Bofors L/60.

45. Bofors L/70 Battery

The AAA values are from *Air Strike*.

Singh (ch. 4) describes the Israeli use.

Sources:

- Wikipedia on the Bofors L/70.
- WeaponSystems on the Bofors L/70.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

46. Bofors L/70 BOFI and BOFI-R Battery

The AAA values are adapted from *Air Strike* values for the Bofors L/70, with the hit rolls increased by one for proximity fuzes, one for the BOFI laser rangefinder (effectively FCR-R), and another one for the BOFI-R radar, giving hit rolls of 5/5/4 for BOFI and 6/6/5 for BOFI-R whereas *Air Strike* gives the BOFI-R 5/5/4.

I don’t know when the BOFI and BOFI-R became available or who else uses them.

Sources:

- Wikipedia on the BOFI and BOFI-R.
- WeaponSystems on the BOFI and BOFI-R.

47. S-60 Battery

The AAA values are from *Air Strike* and *The Speed of Heat*.

Sources:

- Wikipedia on the S-60.
- WeaponSystems on the S-60.

48. ZSU-57-2 Section

For the ZSU-57-2 section: The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the ZSU-57-2.
- WeaponSystems on the ZSU-57-2.

49. M51 Battery

Wikipedia gives a rate of fire of 45 rounds per minute, a muzzle velocity of 854 m/s, an effective maximum altitude of 30k feet, and a horizontal range of 26 hexes (13 km).

The TM states that the search radar is X-band (6–11 GHz or SHF).

The AAA hit rolls are based on the S-60, reduced by 1 for a lower rate of fire, increased by 1 for proximity fuzes, and increased by 2 for FCR-W, so 4/4/3. I will give it the same range of 6/12/18 as the 85 mm JS-12, which fires a slightly heavier round at a slightly lower velocity.

Sources:

- Wikipedia on the M51 and T38 radar
- WeaponSystems on the M51.
- Werrell, K.P. (2005, Air University Press, 2nd edition): “Archie to SAM”.
- Department of the Army (1956): TM 9-6092-1-1 “Antiaircraft Fire Control System M33C and M33D Operation, June 1956”.

50. KS-12 Battery

The AAA values are from *Air Strike* and *The Speed of Heat*.

There are two KS-12 guns. The M-1939 is given a maximum altitude of 34k feet by Wikipedia and 25k feet by Singh. The M-1944 is given a maximum altitude of 38k ft by Wikipedia 29k ft by Singh. The game version have maximum altitudes of 27k feet, which is half-way between the figures given by Singh.

Wikipedia gives maximum horizontal ranges of 15.65 km and 18 km for the M-1939 and M-1944.

Wikipedia gives the rate of fire as 10–12 RPM.

It is not clear to me that the Soviets had proximity fuzes for this gun.

Sources:

- Wikipedia on the KS-12.
- WeaponSystems on the KS-12.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

51. 90 mm M2

The M1 version entered service in 1940. The M2 was a development of the M1 and was the “standard weapon” from 1943. I presume all post-war examples are M2s.

Wikipedia gives the rate of fire as 24 RPM, about twice that of the KS-12.

Wikipedia gives a horizontal range of 19 km, and vertical range of 43.5k feet. Warrell states the effective ceiling is 32k feet. As the calibre and the muzzle velocity of the M2 are between the KS-12 (maximum altitude of 27) and the KS-19 (maximum altitude of 39), this seems to be about right.

I give the M2 the hit rolls of the KS-12 increased by 1, to account for the increased rate of fire and proximity fuzes (although the number of guns per battery is reduced from six to four), a maximum altitude of 32 (following Warrell), and the ranges of the KS-19.

Sources:

- Wikipedia on the 90 mm M2.
- Werrell, K.P. (2005, Air University Press, 2nd edition): “Archie to SAM”.

52. QF 3.7 inch Battery

Wikipedia gives a maximum altitude of 45k feet, a muzzle velocity of 1044 m/s (significantly more than other similar guns), and a rate of fire of 20 rounds per minute.

I give the QF 3.7 the hit rolls of the KS-12 increased by 1, to account for the increased rate of fire and proximity fuzes, a maximum altitude of 45, and a maximum range of 27.

Singh states that by 1956 the IDF had twelve 3.7 inch guns with FCR.

Singh states that Pakistan used the 3.7 inch gun in the 1971 War.

Sources:

- Wikipedia on the QF 3.7 inch.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

53. KS-19 Battery

The AAA values are from *Air Strike*.

Wikipedia gives a rate of fire of 15 RPM, a vertical range of 48 altitude levels (15 km, with a proximity fuse), and a horizontal range of 39 hexes (21 km). The *Air Strike* values are 39 altitude levels and 21 hexes, so are cut back somewhat from the maximums.

It is not clear to me that the Soviets had proximity fuzes for this gun. The one suggestion that they do is that Wikipedia mentions different maximum altitudes for proximity and timed fuzes, but I can find no information on Soviet proximity fuzes.

Sources:

- Wikipedia on the KS-19.
- WeaponSystems on the KS-19.

54. 120 mm M1 Battery

Wikipedia gives rate of fire of 12 RPM, a maximum altitude of 57 altitude levels (17.5 km), and a maximum range of 50 hexes (25 km).

The hit rolls are adapted from the KS-30 (which has a similar rate of fire), with a +1 modifier for proximity shells (although the number of guns per battery is reduced from six to four).

Sources:

- Wikipedia on the M1.

55. KS-30 Battery

Wikipedia gives the rate of fire is 10–12 RPM, the vertical range as 72 altitude levels (22 km) and 54 hexes (29 km) horizontally. Wikipedia also says it provided a 25% increase over the effective ceiling of the KS-19, which would be 48 altitude levels. However, this has a heavier shell and slightly higher muzzle velocity than the 120 mm M1, so it should have a maximum altitude of more than 57. I will give it a maximum altitude of 64.

It is not clear to me that the Soviets had proximity fuzes for this gun.

The hit rolls are adapted from the KS-19 (which has a similar rate of fire). In order to reach an altitude of 64, it needs a maximum range of at least 32. I will give it 36.

Sources:

- Wikipedia on the KS-30.

56. SCR-584

I classify this as FCR-A as the SON-9 is derived from it.

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band and ITU UHF) and the search and tracking ranges for bomber-sized targets as 64 km and 29 km.

Werrell mentions its use with the 90 mm and 3.7 inch guns against V-1s in 1944.

Sources:

- Wikipedia on the SCR-584, 90 mm M1/M2 gun, 120 mm M1 gun, and QF 3.7 inch AA gun.
- Werrell, K.P. (2005, Air University Press, 2nd edition): “Archie to SAM”.

57. SON-9

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band and ITU UHF).

Wikipedia states this was used with the 57, 85, 100, and 130 mm guns and was widely deployed in the Vietnam War. In *The Speed of Heat* the 57 mm S-60 and 85 mm KS-12 are often combined with an FCR-A, presumably the SON-9.

Sources:

- Wikipedia on the SON-9.

58. Super Fledermaus

Wikipedia states there were two versions: the Feuerleitgerät (FLt Gt) 63 (from 1965) and the Feuerleitgerät 69 (from 1970). The latter had “background clutter suppression,” which probably is equivalent to MTI.

Wikipedia states that it can control *two* Oerlikon GDF mounts, hence the restriction to controlling sections and not batteries.

Wikipedia also states these the prototype of the Gepard used a Flt Gt 63. Since the FCR-W radar of the Gepard is equivalent to a +2 modifier and has VF frequency, it corresponds to an FCR-C. We assume the Super Fledermaus does too.

Wikipedia gives the frequency as 8.6–9.6 GHz (NATO X-band and ITU SHF).

The service history is from Wikipedia and Singh (for Israeli use).

Sources:

- Wikipedia on the Super Fledermaus and its use with the Oerlikon GDF.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

59. Flycatcher

Flycatcher is similar to the Cheetah radar, so presumably FCR-C.

According to Wikipedia, the search radar is I/J band (ITU SHF) and the tracking radar is K band (ITU SHF/EHF).

Sources:

- Wikipedia on the Flycatcher.
- Terug naar de toekomst: De Flycatcher.

60. Skyguard

Sources:

- Wikipedia on the Super Fledermaus and its use with the Oerlikon GDF.

61. SA-7A Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- Zaloga gives an in-service date of 1968.
- *Air Strike* gives this a seeker type of I, but that does seem appropriate for an uncooled PbS seeker. Instead, I use a seeker type of E, by analogy with the uncooled seeker in the AIM-9B, and works at about 2 microns.
- *Air Strike* gives a turn rate of HT, but the control capability seems to be the same as in the SA-7B and SA-15. I will adopt BT.
- *Air Strike* gives this a speed of 8. Zaloga gives a lifetime of 8 seconds and an average speed of 1550 km/h. This corresponds to a range of 5.2 km or about 10 hexes.

62. SA-7B Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- *Air Strike* gives this a seeker type of I. That seems about right. The seeker was PbS and cooled electrically, like the AIM-9E which also has a type I seeker.
- *Air Strike* gives a turn rate of BT, which I adopt.
- *Air Strike* gives this a speed of 10. Zaloga states that it has a slightly more powerful rocket motor than the SA-7A, but the mean speed is the same at 1550 km/h. I adopt a speed of 10, like that of the SA-7A.

63. SA-14 Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- The seeker is cryogenically cooled at works at about 4 microns (Zaloga). *Air Strike* gives this a type A seeker, but that does seem appropriate; the first type A seekers in Soviet AAMs did not appear until 1985 in the AA-8C. Also, type A seekers have all-aspect capability Zaloga says that the seeker has similar performance to the RIM-43C, but only limited all-aspect capability. A type M seeker seems to be more appropriate.
- *Air Strike* gives a turn rate of ET, but the control capability seems to be the same as in the SA-7A and SA-7B. I will adopt BT.
- *Air Strike* gives this a speed of 12. Zaloga gives a lifetime of 8 seconds and an average speed of 1440 km/h (as it is slightly heavier than the SA-7). This corresponds to a range of 4.8 km or about 9 hexes.
- The warhead is the same as on the SA-7A/7B.

64. SA-16/18 Squad

The launcher and SAM values are from Malcolm Pipes with significant modifications:

- The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas had heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.
- Carl comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.
- I have reduced the lock-on range of the SA-16 to match that of the SA-14.
- I have increased the flare rating of the SA-16 to match that of the SA-14.
- I have given both missiles the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

65. FIM-43C Squad

The launcher and SAM values are from *Air Strike*.

66. FIM-92A/B/C/D Squad

The launcher and SAM values for the A model are from *Air Strike*. Those for the B/C/D models are from Malcolm Pipes compendium.

- I have increased the flare susceptibility of the FIM-92A from 2 to 3. It did not seem to have significant protection.
- Wikipedia gives an effective range of 5 miles, which is equivalent to a speed of 15. I have left the speed at 14, as that is close enough.
- Wikipedia gives a peak speed of 1930 mp and a self-destruct time of 17 seconds.
- *Air Strike* has the visibility as 2, but looking at photos I don't see much differ

67. Blowpipe Squad

The launcher and SAM values are from *Air Strike*.

68. Javelin/Starburst Squad

The launcher and SAM values for Javelin are from *Air Strike*. The launcher and SAM values for Javelin are just those for Javelin but with the OG guidance replaced by LG guidance. I have seen mention of a Javelin LML; did anyone use it or was it just a prototype?

69. Starstreak Squad

The launcher and SAM values are from Malcolm Pipes. The real darts do not have proximity fuzes, but the missile in the game can obtain a proximity hit. Perhaps a proximity hit here means a hit with only one dart?

70. Starstreak LML Squad

The SAM values are from Malcolm Pipes. I have guessed the launcher values are the same as for shoulder-launched version, just that the launcher has three ready missiles. I don't know if the launcher has reloads. Is the launcher really portable or does it have to be transported by a vehicle?

71. Mistral Squad

The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

72. RBS 70 Squad

For the RBS 70: The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

73. Ayn-al-Saqr Squad

The launcher and SAM values are taken to be equal to those of the SA-7B.

74. British Army Units (Early 1960s)

The British Army formations from early 1960s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

75. The primary source for the orders of battle at Goose Green is Mark Adkin's *The Battle for Goose Green: A Battle is Fought to be Won* (Leo Cooper, 1992).

76. British Army Units (Mid 1980s)

The British Army formations from mid 1980s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

77. Rules

Much of this section should be included in the 3A rules, but it's here for the moment.

78. Squads and Sections as Damaged Platoons

So for example sections/squads have 2D/1D damage resilience and +1/+2 modifiers to AAA attacks. Similar modifiers should be applied to ground-to-ground attacks.

79. Damage Resilience in *Air Strike*

There is a form of damage resilience in *Air Strike*, in that units whose counters have white text are only able to take 1D of damage before being killed (see the last paragraph of rule 28). Such units include infantry SAMs, armored CCUs, and many vehicle SAMs. TSOH also mentions units being able to take only 2D damage, but it doesn't have any relevant units since it attaches SAM teams to infantry platoons rather than having them as separate (see below) and doesn't have a separate FAC unit. In both cases, the idea seems to have been incompletely implemented in the rules. I'm generalizing and homogenizing the idea. This is important as we need sections for mobile AAA units with two vehicles and for the Oerlikon GDF (as Skyguard can only control two guns and not a full

battery). To a large degree, a section is equivalent to a platoon that has taken 1D and a squad is equivalent to a platoon that has taken 2D, so the mechanics of the game are really not impacted.

80. Hard Targets

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8 in *Air Strike*.

81. Open Targets

The BTR-50P APC and the M3 half-tracked APC are examples of open targets. In the 1956 and 1967 Arab-Israeli Wars where both sides used M3 half-tracks. If an M3 is attacked by an air-burst or napalm bomb, the crew and passengers will suffer more or less as if they were in the open.

82. Soft Targets

Examples of soft targets include infantry, towed artillery, unarmored vehicles, and armored vehicles that do not fully protect their personnel and/or weapons such as the M-107 and M-110 tracked artillery vehicles.

Nothing like the M-107 or M-110 is included in *Air Strike*, but again I would assert that it is tracked mainly for mobility. Effectively, these are towed artillery installed in an exposed position on a tracked hull and have almost no protection for most of the crew.

Most tracked SAM vehicles are classified as hard targets in *Air Strike* and *Eagles of the Gulf*, despite many of them having completely exposed missiles. If an air-burst bomb explodes above a SAM-4 launcher, it is not going to shrug it off. I think many SAM vehicles are tracked mainly for *mobility* and perhaps for crew protection, but in terms of a mission-kill on their weapon systems, they are soft targets. I've not changed anything yet, but we need to think about this.

83. Representation

The representation is based on NATO symbols, including those for infantry, armor, anti-armor, air defense, guns, missiles, mortar, tube artillery, and rocket artillery. One addition is that a rectangle denotes an unarmored vehicle such as a truck or tracked carrier. A truck or mobile unit has two wheel symbols in the lower position, whereas a tracked carrier has a tracked symbol. The mortar symbol is modified to be simply an open circle, for compactness.

All vehicular units have either an armored or unarmored vehicle symbol, all infantry units have an infantry symbol, and towed units have none of these.

The letter T in the central position indicates a transport capacity (e.g., APCs, IFVs, cargo trucks, and tracked cargo carriers). Words and letters in the lower position indicate variants on a basic type: the letters H, M, L, and O on an armored unit indicate heavy, medium, light, and open armored protection; the word WPN or FAC on an infantry unit indicates a weapons or FAC unit; and the letter H on an infantry SAM unit indicates a heavy version of the launcher (e.g., the LML version of Starstreak).

84. Mobility

The inspiration for this rule was this photograph of the Mitla Pass from 1967, which shows trucks destroyed on and just off a road. Also, in the the Falklands War roads could be traversed by Landrovers and AMLs, but off-road terrain was impassable to everything except the CVR(T)s and Bv 202s despite being open.

85. Mobility Restrictions

When trucks have good mobility, the rule is: don't put trucks or towed weapons in forests unless there is a road. When trucks have poor mobility, the rule is effectively: if the terrain is difficult, keep your trucks and towed weapons near roads even in clear terrain.

86. Truck Mobility

For example, a scenario set in the open desert of southern Iraq might specify that trucks have good mobility, but one set in the Falkland Islands might specify that they have poor mobility. Similarly, a scenario might specify that cargo trucks have poor mobility, but mobile artillery units have good mobility. Finally, a scenario might not assign an explicit transport unit to a towed artillery battery, but might state that it has been transported by helicopter and so can be placed as if it had a transport unit with good mobility.

87. Modifying Mobility Classes

One common modification is to explicitly indicate the hexes occupied by each ground unit; in this case this rule is largely superfluous (although if we allow ground units to move, then the restrictions can still be relevant).

Panhard AMLs are wheeled armored vehicles and so normally have unrestricted mobility. Nevertheless, in the Falklands War they were restricted to

roads and trails, and this corresponds to poor mobility. A scenario could fix this problem by explicitly giving them poor mobility.

88. Composite Platoons

This rule comes from TSOH. It doesn't have counters for infantry SAMs, so in scenarios V-11 and V-22 it says treat infantry platoons as if they have an infantry SAM squad. This rule helps with stacking (since attached squads do not count for stacking) and with keeping your precious FACs and SAM teams anonymous. It also works nicely with the transport rule, since attached squads do not need their own transport but can ride with the platoon.

89. Damage to Attached Squads

The roll of 4+ seems about right. The probability of killing an attached squad is 40% for 1D and 64% for 2D. Those numbers are fairly good approximations to 1/3 and 2/3. If the roll was 3+, they would be 30% and 51%, and that seems a little low.

Are the damage rolls secret?

90. Transportation

The idea of transporting units and cargo comes from the "Thunder and Fire" scenario from *Eagles of the Gulf*. The rules here simply generalize it slightly, incorporate the idea of mobility, and add rules for mounting and dismounting.

91. Mobility

Being placed according to the mobility of its transport allows a tracked transport to place a towed unit in a forest hex.

92. Mounting and Dismounting Infantry

The rules for mounting and dismounting are completely new. I'm very open to suggestions about this.